

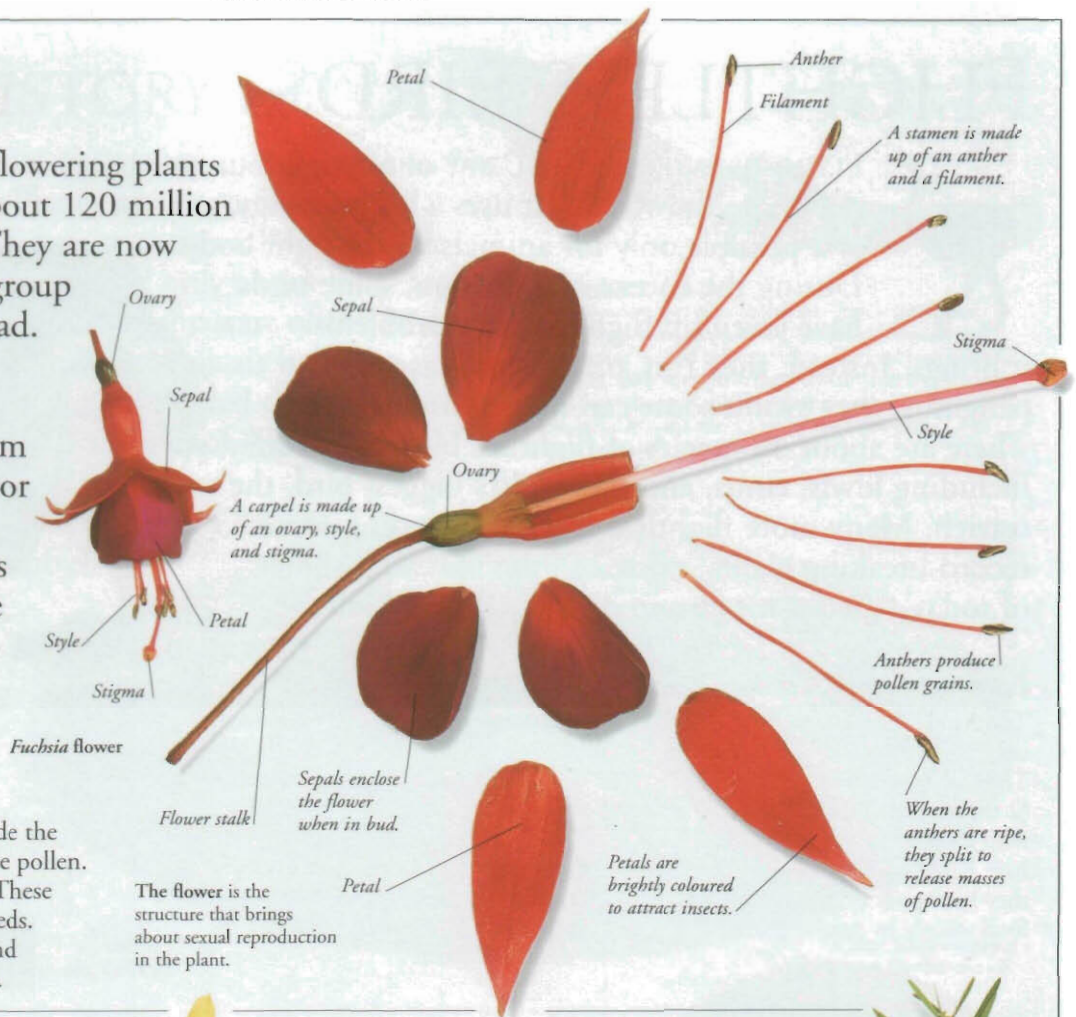
FLOWERS



THE FIRST flowering plants appeared about 120 million years ago. They are now the largest group of plants, and are widespread. Flowers are the advanced reproductive structures of plants. The majority of them are pollinated by the wind or by animals. Over millions of years, flowers and insects have co-evolved to produce some very complex and interesting relationships.

Parts of a flower

A plant's reproduction organs are inside the flowers. Stamens are male and produce pollen. The female organs are called carpels. These contain ovules, which develop into seeds. A ring of petals attracts pollinators, and sepals protect the flower when in bud.



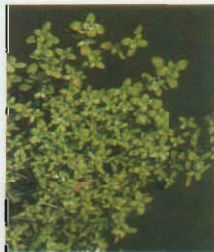
Pollination

In order for seeds to develop, a flower has to be pollinated. Pollen from the stamens of a flower of the same species must stick to the plant's stigma. Cross-pollination occurs when pollen from one plant lands on the flowers of another. If a flower is pollinated by its own pollen, it is called self-pollination.

Water pollination

Aquatic plants may have aerial, submerged, or floating flowers. A few use the surface film of water to carry pollen. The flowers float in shallow dips. The pollen slides into these and pollinates the flowers.

Starwort



Bird pollination

Many flowers that rely on birds to pollinate them are red or orange – colours that attract birds. The flowers tend to be tubular so that birds can dip their beaks in. Stamens dust the birds' heads with sticky pollen.



Hummingbird at a thistle flower



Bee at a narcissus flower

Insect pollination

Flowers are mostly pollinated by insects. These are attracted to the flowers by their petals, a scent, and a supply of nectar or pollen to eat. The pollen sticks to a visiting insect and is carried to another flower.

Wind pollination

Some plants rely on wind to waft their pollen from one flower to another. Their flowers may have no petals, or just tiny ones. They are often arranged in catkins with conspicuous stamens and stigmas.

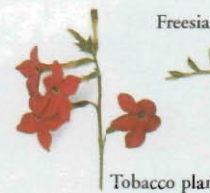
Hazel catkins and pollen



Mammal pollination

Important mammal pollinators include some species of tropical bat and many types of tiny Australian possum. They pollinate flowers as they feed on nectar and pollen.

Pygmy possum feeding on a Banksia flower



Freesia

Tobacco plant

Scent and colour

The colour and scent of a flower attract insects or other animals. Drops of sugary nectar secreted at the base of the petals provide food for the insects, which are dusted with pollen while they drink. Flowers are often sweetly scented, but some smell unpleasant, especially those pollinated by flies.



Wattle

Insect mimics

Many orchids have such specialized partnerships that only one type of insect serves as a pollinator. Some orchids look and smell so like a female insect, that males of that species try to mate with them, picking up pollen as they do so.

Bee orchid

The bee orchid can be pollinated by a bee, but is often self-pollinated by the wind.



An orchid's pollen is produced in small clumps called pollinia.

Part of the flower looks and smells just like a bee.

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